

JOSHUA D. BRAKENSIEK, PH.D.

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Curriculum Vitae

Positions	UNIVERSITY OF CALIFORNIA, BERKELEY Postdoctoral Scholar in EECS Supervisor: Venkatesan Guruswami	2024 –
Education	STANFORD UNIVERSITY Doctor of Philosophy in Computer Science Co-advisors: Aviad Rubinstein and Moses Charikar Thesis: “Higher Order MDS Codes”	2018 – 2024
	STANFORD UNIVERSITY Master’s of Science in Computer Science Qualification Exam Topic: “Automated Design of Error-Correcting Codes”	2018 – 2021
	CARNEGIE MELLON UNIVERSITY Master’s of Science in Mathematical Sciences Thesis: “Polymorphic Inquiries: Promise Constraint Satisfaction and Beyond” Advisor: Venkatesan Guruswami	2016 – 2018
	CARNEGIE MELLON UNIVERSITY Bachelor’s of Science in Mathematical Sciences Minor in Science, Technology, and Society University and Mellon College of Science Honors GPA: 4.00/4.00	2014 – 2018
Publications	REDUNDANCY IS ALL YOU NEED Brakensiek, J. and Guruswami, V. Symposium on Theory of Computing (STOC), 2025. arXiv:2411.03451	
	ON THE MYSTERIES OF MAX NAE-SAT Brakensiek, J. , Huang, N., Potechin, A., and Zwick, U. SIAM Journal on Discrete Mathematics (SIDMA), 2025. arXiv:2009.10677 Conference Version: Symposium on Discrete Algorithms (SODA), 2021.	
	GENERIC REED-SOLOMON CODES ACHIEVE LIST-DECODING CAPACITY Brakensiek, J. , Gopi, S., and Makam, V. SIAM Journal on Computing 2024. arXiv:2206.05256 Special Issue. Conference Version: Symposium on Theory of Computing (STOC), 2023.	

IMPROVED FIELD SIZE BOUNDS FOR HIGHER ORDER MDS CODES

Brakensiek, J., Dhar, M., and Gopi, S.

IEEE Transactions on Information Theory, 2024. arXiv:2212.11262

Conference Version:

IEEE International Symposium on Information Theory (ISIT), 2023.

ROBUST FACTORIZATIONS AND COLORINGS OF TENSOR GRAPHS

Brakensiek, J. and Davies, S.

SIAM Journal on Discrete Mathematics (SIDMA), 2024. arXiv:2207.08913

AG CODES ACHIEVE LIST DECODING CAPACITY OVER CONSTANT-SIZED FIELDS

Brakensiek, J., Dhar, M., Gopi, S., and Zhang, Z.

Symposium on Theory of Computing (STOC), 2024. arXiv:2310.12898

GENERALIZED GM-MDS: POLYNOMIAL CODES ARE HIGHER ORDER MDS

Brakensiek, J., Dhar, M., and Gopi, S.

Symposium on Theory of Computing (STOC), 2024. arXiv:2310.12888

TIGHT APPROXIMABILITY OF MAX 2-SAT AND RELATIVES, UNDER UGC

Brakensiek, J., Huang, N., and Zwick, U.

Symposium on Discrete Algorithms (SODA), 2024. arXiv:2310.12911

SEPARATING MAX 2-AND, MAX DI-CUT AND MAX CUT

Brakensiek, J., Huang, N., Potechin, A., and Zwick, U.

Symposium on Foundations of Computer Science (FOCS), 2023. arXiv:2212.11191

Invited to SICOMP Special Issue.

A DICTATORSHIP TEST WITH PERFECT COMPLETENESS FOR 2-TO-2 LABEL COVER

Brakensiek, J. and Guruswami, V.

Chicago Journal of Theoretical Computer Science (CJTCS), 2023. ECCC TR17-141

SDPs AND ROBUST SATISFIABILITY OF PROMISE CSP

Brakensiek, J., Guruswami V., and Sandeep, S.

Symposium on Theory of Computing (STOC), 2023. arXiv:2211.08373

CONDITIONAL DICHOTOMY OF BOOLEAN ORDERED PROMISE CSPs

Brakensiek, J., Guruswami V., and Sandeep, S.

TheoretCS, 2023. arXiv:2102.11854

Conference Version:

Intl. Colloquium on Automata, Languages, and Programming (ICALP) 2021.

LOWER BOUNDS FOR MAXIMALLY RECOVERABLE TENSOR CODES AND HIGHER ORDER MDS CODES

Brakensiek, J., Gopi, S., and Makam V.

IEEE Transactions on Information Theory, 2022. arXiv:2107.10822

THE RESOLUTION OF KELLER'S CONJECTURE

Brakensiek, J., Heule, M., Mackey, J., and Narváez, D.

Journal of Automated Reasoning (JAR), 2022. arXiv:1910.03740

Special Issue.

Conference Version:

International Joint Conference on Automated Reasoning (IJCAR), 2020

Best Paper Award.

CONSTRAINT SATISFACTION PROBLEMS WITH GLOBAL MODULAR CONSTRAINTS: ALGORITHMS AND HARDNESS VIA POLYNOMIAL REPRESENTATIONS

Brakensiek, J., Gopi, S., and Guruswami, V.

SIAM Journal on Computing (SICOMP), 2022. arXiv:1902.04740

Conference Version:

Symposium on Theory of Computing (STOC), 2019.

THE QUEST FOR STRONG INAPPROXIMABILITY RESULTS WITH PERFECT COMPLETENESS

Brakensiek, J. and Guruswami, V.

ACM Transactions on Algorithms (TALG), 2021. ECCC TR17-80

Conference Version:

Intl. Workshop on Approximation Algorithms for Combinatorial Optimization Problems (APPROX) 2017.

PROMISE CONSTRAINT SATISFACTION:

ALGEBRAIC STRUCTURE AND A SYMMETRIC BOOLEAN DICHOTOMY

Brakensiek, J. and Guruswami, V.

SIAM Journal on Computing (SICOMP), 2021. arXiv:1704.01937

Conference Version:

Symposium on Discrete Algorithms (SODA), 2018.

THE POWER OF THE COMBINED BASIC LP AND AFFINE RELAXATION FOR PROMISE CSPs.

Brakensiek, J., Guruswami, V., Wrochna, M., and Živný S.

SIAM Journal on Computing (SICOMP), 2020. arXiv:1907.04383

Conference version:

SYMMETRIC POLYMORPHISMS AND EFFICIENT DECIDABILITY OF PROMISE CSPs.

Brakensiek, J. and Guruswami, V.

Symposium on Discrete Algorithms (SODA), 2020.

SMOOTHED COMPLEXITY OF 2-PLAYER NASH EQUILIBRIA

Boodaghians, S., **Brakensiek, J.**, Hopkins, S., and Rubinstein, A.

Symposium on Foundations of Computer Science (FOCS), 2020. arXiv:2007.10857

CODED TRACE RECONSTRUCTION IN A CONSTANT NUMBER OF TRACES.

Brakensiek, J., Li, R., and Spang, B.

Symposium on Foundations of Computer Science (FOCS), 2020. arXiv:1908.03996

CONSTANT-FACTOR APPROXIMATION OF NEAR-LINEAR EDIT DISTANCE IN NEAR-LINEAR TIME.

Brakensiek, J. and Rubinstein, A.

Symposium on Theory of Computing (STOC), 2020. arXiv:1904.05390

BRIDGING BETWEEN 0/1 AND LINEAR PROGRAMMING VIA RANDOM WALKS

Brakensiek, J. and Guruswami, V.

Symposium on Theory of Computing (STOC), 2019. arXiv:1904.04860

AN ALGORITHMIC BLEND OF LPs AND RING EQUATIONS FOR PROMISE CSPs

Brakensiek, J. and Guruswami, V.

Symposium on Discrete Algorithms (SODA), 2019. ECCC TR18-059

EFFICIENT LOW-REDUNDANCY CODES FOR CORRECTING MULTIPLE DELETIONS

Brakensiek, J., Guruswami, V., and Zbarsky, S.

IEEE Transactions on Information Theory (ITIT), 2017. arXiv:1507.06175

Conference version:

Symposium on Discrete Algorithms (SODA) 2016.

VERTEX ISOPERIMETRY AND INDEPENDENT SET STABILITY FOR TENSOR POWERS OF CLIQUES

Brakensiek, J.

Intl. Workshop on Randomization and Computation (RANDOM), 2017. arXiv:1702.04432

NEW HARDNESS RESULTS FOR GRAPH AND HYPERGRAPH COLORINGS

Brakensiek, J. and Guruswami, V.

Computational Complexity Conference (CCC), 2016. ECCC TR16-029

EFFICIENT GEOMETRIC PROBABILITIES OF

MULTI-TRANSITING EXOPLANETARY SYSTEMS FROM CORBITS

Brakensiek, J., and Ragozzine, D.

The Astrophysical Journal, 2016. arXiv:1602.07014, Open source CORBITS code.

Manuscripts RIGIDITY MATROIDS AND LINEAR ALGEBRAIC MATROIDS WITH APPLICATIONS TO MATRIX COMPLETION AND TENSOR CODES

Brakensiek, J., Dhar, M., Gao, J., Gopi, S., Larson, M. arXiv:2405.00778

KAPRANOV DEGREES

Brakensiek, J., Eur, C., Larson, M., and Li, S. arXiv:2308.12285

A SIMPLE SUBLINEAR ALGORITHM FOR GAP EDIT DISTANCE

Brakensiek, J., Charikar, M., and Rubinfeld, A. arXiv:2007.14368

BOUNDS ON THE SIZE OF SOUND MONOTONE SWITCHING

NETWORKS ACCEPTING PERMUTATION SETS OF DIRECTED TREES

Brakensiek, J., and Potechin, A. arXiv:1301.3780

**Other
Writings**

AUTOMATED DESIGN OF ERROR-CORRECTING CODES PART 2

Theory Dish (Stanford's CS Theory Research Blog). Wordpress.

AUTOMATED DESIGN OF ERROR-CORRECTING CODES PART 1

Theory Dish (Stanford's CS Theory Research Blog). Wordpress.

**Research
Talks and
Posters**

INTERNATIONAL SYMPOSIUM ON INFORMATION THEORY
Hybrid. June 2023.

DAGSTUHL SEMINAR 22201
Hybrid. May 2022. (two talks)

STANFORD TCS QUALIFICATION EXAM
Virtual. March 2021.

STANFORD THEORY LUNCH
Virtual. December 2020.

FOUNDATIONS OF COMPUTER SCIENCE
Virtual. October 2020.

MICROSOFT RESEARCH MLO SEMINAR (INVITED)
Virtual. September 2020.

CSP SEMINAR (INVITED)
Virtual. August 2020.

SYMPOSIUM ON THEORY OF COMPUTING
Virtual. June 2020.

SYMPOSIUM ON DISCRETE ALGORITHMS
Salt Lake City, Utah, January 2020.

ACO SEMINAR
Carnegie Mellon University, Pittsburgh, Pennsylvania, September 2019.

MICROSOFT RESEARCH
Redmond, Washington, July 2019.

SYMPOSIUM ON THEORY OF COMPUTING
Phoenix, Arizona, June 2019. (talk and poster)

STANFORD THEORY LUNCH
Stanford, California, January 2019.

SYMPOSIUM ON DISCRETE ALGORITHMS
San Diego, California, January 2019.

DAGSTUHL SEMINAR 18231
Schloss Dagstuhl, Wadern, Germany, June 2018
Recipient of NSF Support Grant

CMU THEORY LUNCH (INVITED TALK)
Carnegie Mellon University, Pittsburgh, Pennsylvania, January 2018.

AMS CONTRIBUTED PAPERS IN COMBINATORICS
Joint Mathematics Meetings, San Diego, California, January 2018

SYMPOSIUM ON DISCRETE ALGORITHMS

New Orleans, Louisiana, January 2018

APPROX/RANDOM CONFERENCE

Berkeley, California, August 2017 (two talks)

CONFERENCE ON COMPUTATIONAL COMPLEXITY

Tokyo, Japan, May/June 2016

MEETING OF THE MINDS UNDERGRADUATE POSTER SESSION

Carnegie Mellon University, Pittsburgh, Pennsylvania, May 2016

Recipient of Early Research Award

CMU THEORY LUNCH (INVITED TALK)

Carnegie Mellon University, Pittsburgh, Pennsylvania, January 2016.

SYMPOSIUM ON DISCRETE ALGORITHMS

Arlington, Virginia, January 2016.

CMU THEORY LUNCH

Carnegie Mellon University, Pittsburgh, Pennsylvania, November 2015

PI MU EPSILON UNDERGRADUATE STUDENT PRESENTATIONS

MAA MathFest, Hartford, Connecticut, August 2013

Recipient of Pi Mu Epsilon Student Presentation Award

MAA UNDERGRADUATE POSTER SESSION

Joint Mathematics Meetings, San Diego, California, January 2013

Recipient of Outstanding Presentation Award

DIVISION OF PLANETARY SCIENCE POSTER SESSION

Reno, Nevada, October 2012

Recipient of Hartmann Student Travel Grant

RESEARCH SCIENCE INSTITUTE SYMPOSIUM

MIT, Cambridge, Massachusetts, August 2012

Research Awards

NSF Mathematical Sciences Postdoctoral Fellowship, 2025

FOCS 2023 SICOMP Special Issue Invitation, 2023

STOC 2023 SICOMP Special Issue Invitation, 2023

Microsoft Research PhD Fellowship, 2021

IJCAR Best Paper Award co-winner, 2020

IJCAR 2020 JAR Special Issue Invitation, 2020

NSF Graduate Research Fellowship, 2018

CRA Outstanding Undergraduate Researcher Award, 2018

Goldwater Scholarship, 2016

Davidson Fellow, 2013

Top Five Awardee: Written Research, Research Science Institute, 2012

Competition	Putnam Fellow (top 5 individual) and 1st place team, 2016
Awards	8th place individual and 2nd place team, Putnam Competition, 2015 10.5th place individual, Putnam Competition, 2014 Two-time Gold Medalist: International Olympiad in Informatics, 2013-14 Silver Medalist: International Mathematical Olympiad, 2014 Samuel L. Greitzer/Murray S. Klamkin Award for Mathematical Excellence, 2014 USA Mathematical Olympiad sole perfect scorer, 2014 Akamai Foundation Scholarship, 2014 USA Mathematical Olympiad Winner, 2012, 2014 International Mathematical Olympiad invitee (declined), 2012 Bronze Medalist: Romanian Masters of Mathematics, 2012
Other Awards	CMU Senior Leadership Recognition, 2018 Phi Kappa Phi, 2018 Phi Beta Kappa <i>early inductee</i> , 2017 Knaster-McWilliams Scholar, 2014-18 Pi Mu Epsilon, 2012
Work Experience	COURSE ASSISTANT Spring 2021 Course assistant for Stanford course CS 269I (Incentives in Computer Science) taught by Aviad Rubinstein. <i>Recognized to be in top 5% of CAs.</i> RESEARCH INTERN FOR MICROSOFT Summer 2020 Research in the algorithms group at Microsoft Research under the supervision of Sivakanth Gopi. COURSE ASSISTANT Spring 2019 Course assistant for Stanford course CS 354 (Unfulfilled Algorithmic Fantasies) taught by Aviad Rubinstein. <i>Recognized to be in top 5% of CAs.</i> TEACHING ASSISTANT Fall 2017 Teaching assistant for CMU course 15-458/15-858 (Discrete Differential Geometry) taught by Keenan Crane. RESEARCH ASSISTANT 2015–18 Theoretical computer science research under Venkatesan Guruswami at CMU MATHEMATICS OLYMPIAD GRADER 2014–17 Graded exams which helped to decide the USA International Math Olympiad team RESEARCH ASSISTANT 2015–16 Astrostatistics research under Chad Schafer and Peter Freeman at CMU TECHNICAL CONSULTANT FOR EXPII, INC. 2014 Web design

Other

MCS COLLEGE COUNCIL

Undergraduate representative for 2017-18 academic year.

(SUB)REVIEWER/REFEREE

Conferences: ISIT 2019; ICALP 2019; FOCS 2019; ESA 2019; FSTTCS 2019; SODA 2020; ISIT 2020; ICALP 2020; CCC 2020; MFCS 2020; SODA 2021; APPROX 2021; ICALP 2021; ISIT 2021; MFCS 2021; STOC 2021; FOCS 2021; SODA 2022; CCC 2022; STOC 2022; ICALP 2022; FOCS 2023; ESA 2023; IPCO 2023; SODA 2024; LICS 2024; STOC 2024; ICALP 2024; IWOCA 2024; FOCS 2024; SODA 2025; STACS 2025; STOC 2025; ICALP 2025

Journals: Journal of the ACM; IEEE Transactions on Information Theory; ACM Transactions on Algorithms; SIAM Journal on Computing; Theory of Computing; ACM Transactions on Computation Theory; Electronic Journal of Combinatorics; Theory of Computing Systems; Information and Computation; Complexity; Mathematical Reviews/MathSciNet

PHOTOGRAPHY

Photograph “Symphony of Architectural Geometry” selected as the 2013, Week 52 photo in the MAA Found Math column of the Mathematical Association of America. Also featured in the MAA’s 100th anniversary calendar.